MEDGRID SENTINEL

Intelligent Hospital Microgrids · Predictive Energy Resilience · Uninterrupted Critical Care

One-page summary · Anneru IAS Global Energy Hackathon 2026 (Oct 30 – Nov 1) · **Brief #13 — Power for a city the grid can't protect** · **Track 2** Digital Solutions & AI · **Setting B** Non-residential (healthcare) · Team: [CONFIRM AT SUBMISSION]

Community & target user — who, specifically, and who pays

Urban Nigerian hospitals on feeders that fail with the national grid (250-bed reference hospital, illustrative). User: the facility/works engineer and duty manager. Payer: the hospital, or an energy-service provider under Energy-as-a-Service — the structure of the real NSIA-LUTH cancer-centre microgrid (300 kWp/1.1 MWh, 2025).

The problem — what fails today, and why

Nigeria's grid failed 12+ times in 2024 (industry counts; NERC official count lower). Each collapse drops hospitals onto diesel at ≈7× grid cost (≈■1,560/L vs ≈■210/kWh) and threatens operating theatres, ICU and the vaccine cold chain. Solar+storage is bought blind: no tool shows whether a design holds critical loads through a real outage, or what each avoided diesel litre costs. Root cause: decision quality, not hardware.

Proposed solution — what we built and how it works

A deterministic hospital-microgrid digital twin: a 15-minute-step simulation of PV, battery, grid and genset dispatching around tiered clinical loads (life-critical / clinical-priority / deferrable), with outage islanding and honest deficit reporting — plus a ten-scenario stress lab, an EaaS financial engine, a dual-metric sizing optimizer and explainable predictive intelligence. No black box touches dispatch; the core runs offline with no external APIs.

Proof of Concept for the 48 hours

The full application built during the sprint: deterministic engine (power balance verified every step), 58 unit tests with hand-calculated values, production build verified, results reproducible via `npm run results`.

Brief success metrics — baseline vs Sentinel (from results/demo-results.json, reproducible)

Brief metric	Diesel-only baseline	Sentinel-managed
Critical-service hours (12.5 h islanded window)	12.50 h (diesel-carried)	**12.50 h — 100%, 0.00 kWh unserved Tier-1**
Diesel litres (3-day scenario window)	1,295 L	**763 L — 532 L avoided (–41%)**
Annualized (320 normal + 45 outage days)	58,255 L/yr · ■569.6M/yr opex	**34,345 L/yr · ■457.8M/yr · –■111.8M/yr · –64.1 t CO■/yr**

Estimated cost & who pays — can the lowest-income users afford it?

Reference design CapEx ≈ **■760.4M ($491k)**: payback **7.9 yr**, IRR 4.6%, NPV negative at an 18% commercial discount rate — stated plainly. EaaS fee ≈ ■161M/yr vs ■112M/yr operating savings: closes below a computed **4.6% financing rate** — exactly the gap REA-type/DFI programmes exist to fill; service cost ≈ **■1,764/bed-day**. No grant is assumed to exist anywhere in the model.

6–18-month deployment path — what gets built, by whom

M1–3: metering-first pilot on one real feeder (replace illustrative data). M4–6: engineering design, permits, financing & O&M; agreements with an energy-service partner. M7–12: pilot deployed; measured-vs-predicted evidence. M13–18: replicate to 3–5 facilities; provider playbook. Installation is performed by licensed partners — Sentinel is decision support, never a control system.

How it scales

One facility = one configuration object (loads, tariffs, outage pattern, climate); the rural-clinic profile runs on the same engine today. The sizing optimizer proves designs must match outage timing: daytime outages → PV-first (no battery optimal); evening outages → storage-heavy. Country adaptation = parameter changes (tariffs, diesel/FX, standards, financing), not code changes.

Biggest risk & how we test it

Simulated facility data could mislead the sizing recommendation. Mitigation is structural: every input is labelled, the engine accepts measured AMI data unchanged, and the roadmap is metering-first — month 1 delivers the feeder's real outage pattern before any CapEx decision. Secondary risk: brief definitions pending full kick-off text (logged for reconciliation; metrics computed generically).

Prior-work disclosure

Everything was built during the 48-hour sprint (repository starts at the sprint's opening commit). Tools: AI coding agent as development instrument (disclosed), standard open-source libraries, no external datasets or APIs. Full statement: submission/DISCLOSURE_STATEMENT.txt.